Essential Linux Commands Every Junior Engineer Needs

Linux powers almost all modern cloud servers and DevOps pipelines today. Master these core commands to confidently inspect logs, troubleshoot system crashes, and fix server issues fast.

What you'll learn:

1. Searching Log Files with grep
2. Live Log Streaming with tail
3. Disk Space Inspection with df and du
4. Managing Permissions with chmod

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1

TIP 1 OF 4

Searching Log Files with grep

The grep tool searches through text files line by line to find specific keywords or patterns. Adding the -i flag ignores case sensitivity, while -r searches through entire directories. You can also pipe output from other commands directly into grep to filter results.

```
$ grep -rn "ERROR" /var/log/nginx/
```

2

TIP 2 OF 4

Live Log Streaming with tail

The tail command displays the final lines of a file instead of opening the whole document. Adding the -f flag keeps the terminal open to stream fresh lines continuously as your application writes them. This is essential for live debugging during traffic spikes.

```
$ tail -n 100 -f /var/log/app/production.log
```

3

TIP 3 OF 4

Disk Space Inspection with df and du

Running df -h shows overall storage usage across all mounted drives in human-readable units like MB and GB. If a drive is full, navigate to that folder and run du -sh * to see exactly which files or subfolders are hogging the disk space.

```
$df -h && du -sh /var/log/*
```

4

TIP 4 OF 4

Managing Permissions with chmod

Linux assigns Read (4), Write (2), and Execute (1) permissions to owners, groups, and users. When you create a shell script, it lacks execute permissions by default. Use chmod +x to make the script executable so you can run it directly.

```
$ chmod +x deploy-script.sh
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

- + Always include the -h flag on disk commands to read sizes in human-friendly formats.
- + Pipe large outputs into 'less' (e.g., command | less) so you can scroll up and down easily.
- + Use Ctrl + R in your terminal to quickly search and reuse previously typed commands.
- + Always verify your current working directory with 'pwd' before deleting or moving files.
- + Practice keyboard shortcuts like Ctrl + C to cancel running tasks and Ctrl + L to clear the screen.

Watch Out For

- ! Running recursive delete commands like 'rm -rf' without checking the target directory path.
- ! Forgetting that Linux file names and commands are strictly case-sensitive.
- ! Using 'sudo' unnecessarily for every command instead of setting proper file permissions.
- ! Leaving continuous tail processes running indefinitely on active production servers.

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